Configurational Coupled Cluster for Planar Rotors

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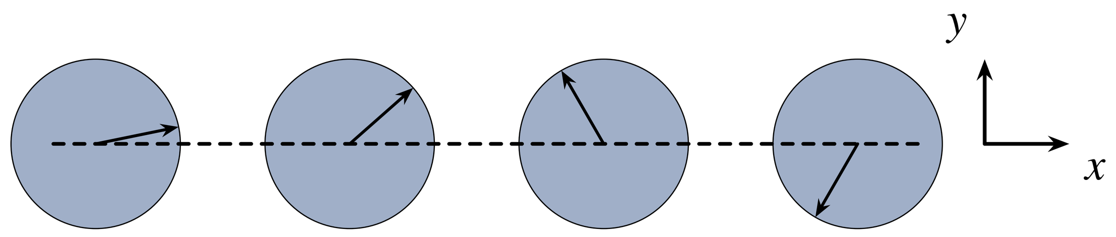
Premise

We would like to use the time idependent Schrodinger Equation to extract eigenvalues and eigenvectors.

$$\hat{H} = B\hat{K} + g\hat{V} \tag{1}$$

Where hamiltonian describes a coplanar rotor system.



We solve Shrodinger equation by diagnilizing hamiltonian:

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle \tag{2}$$

However thise hamiltonians becove extreamly large really fast which makes exact diagnilization computationally heavy. *(Even though we are curently working with systems of rotors the method is equaly aplicable to magnetic systems.)*

Problem Statement

Generalize Coupled Cluster methods (CC) to rotor systems, to explore larger systems. Compare with other approaches and also consider Thermal Properties.

Improving Exact Diagonilization

In order to to explore larger systems with Exact Diagnilization and Configurational Coupled Cluster (CCC) we use Natural Orbitals:

- Truncate the basis using Natural Orbital (NO) density matrix for small system(2-3 sites), and use selected eivencectors.
- Use sparse matrix techniquesto COnstruct Hamiltonian and use direct diagnilization.
- Systems up to about 9 sites with 3-5 NO's can be studied.

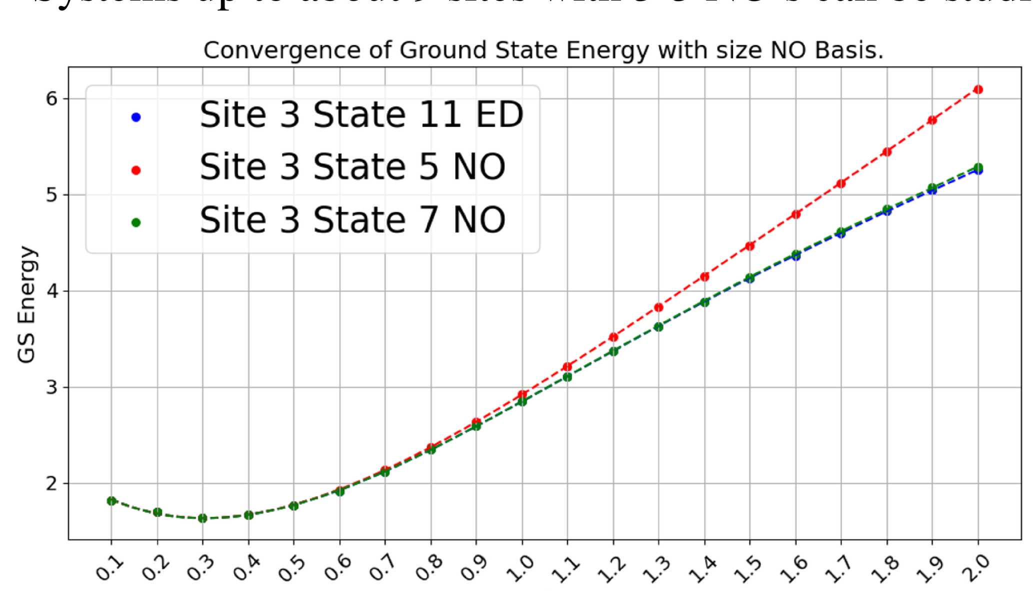


Figure 1: Qualitative demonstration works through the NO truncation method but acuracy decreases with the increase of the coupling strengs.

Periodic vs. Non-Periodic System Setup

CCC calcualations can be sued for both periodic and finite systems. One could go beyond nearest neighbor interactions. The example of the periodic system that we would consider:

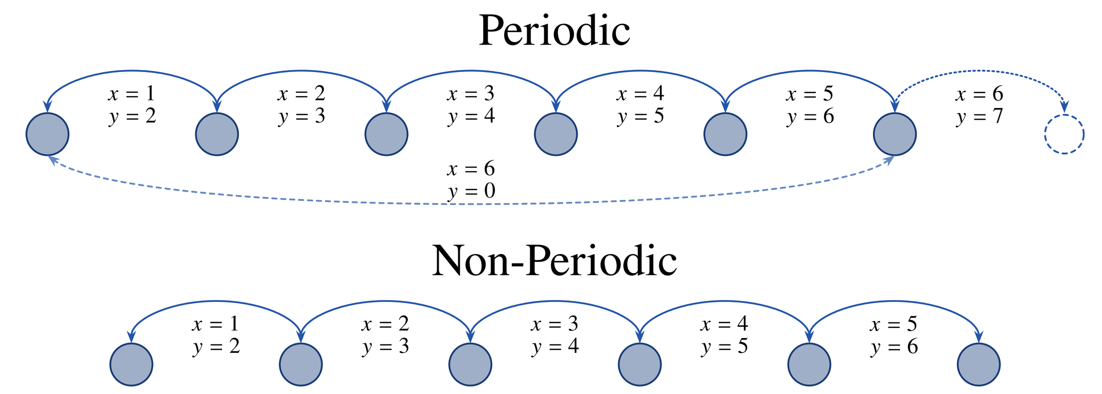


Figure 2: Circles representing rotor sites, x and y respresenting site interactions between sites 1 and 2; 2 and 3: etc. The doted lines and sites are two ways how we can consider our periodic systems

Configurational Coupled Cluster

- CI CCC comparison emfesising the separated exictation operators and ability of CCC to axes higher excitations.

Large $g >= 100$ Limit

One of the properties which we utilise to explore the large g limit ($g >= 100$) is that as g aproaches infininty the system becomes a superposition of two states (alignments) 0 and π.

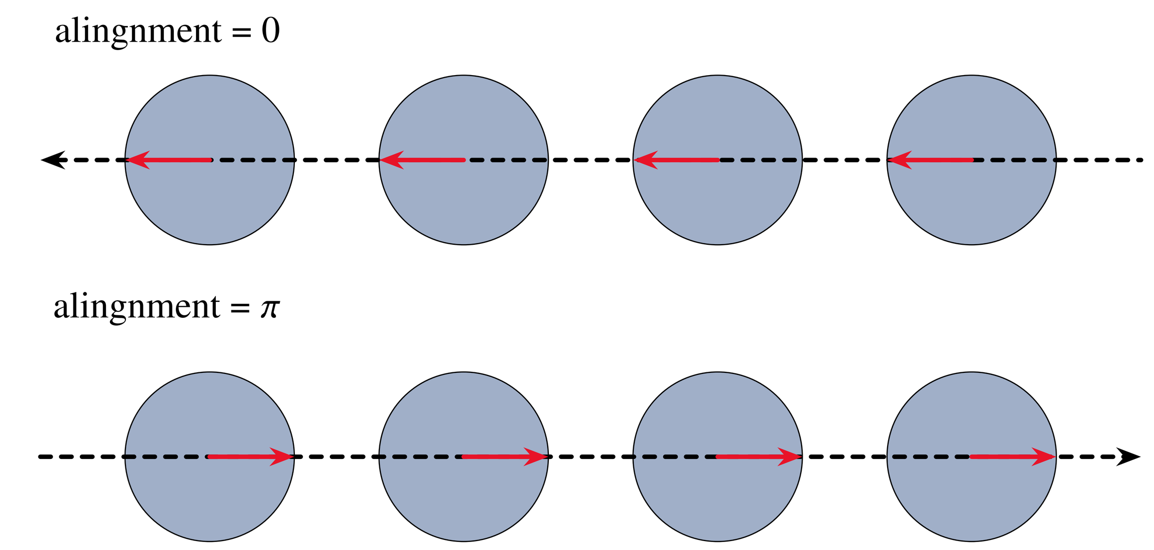


Figure 3: The shcematic for superposition of the two state alignments.

With g large we can the wavefunctions assiciated with 0 and π alignment are separate. This property allows us to isolate for one of them in the following maner by taking a linear combination of the 0 and π wavefunctions and constructing a Natural orbital basis based on them.

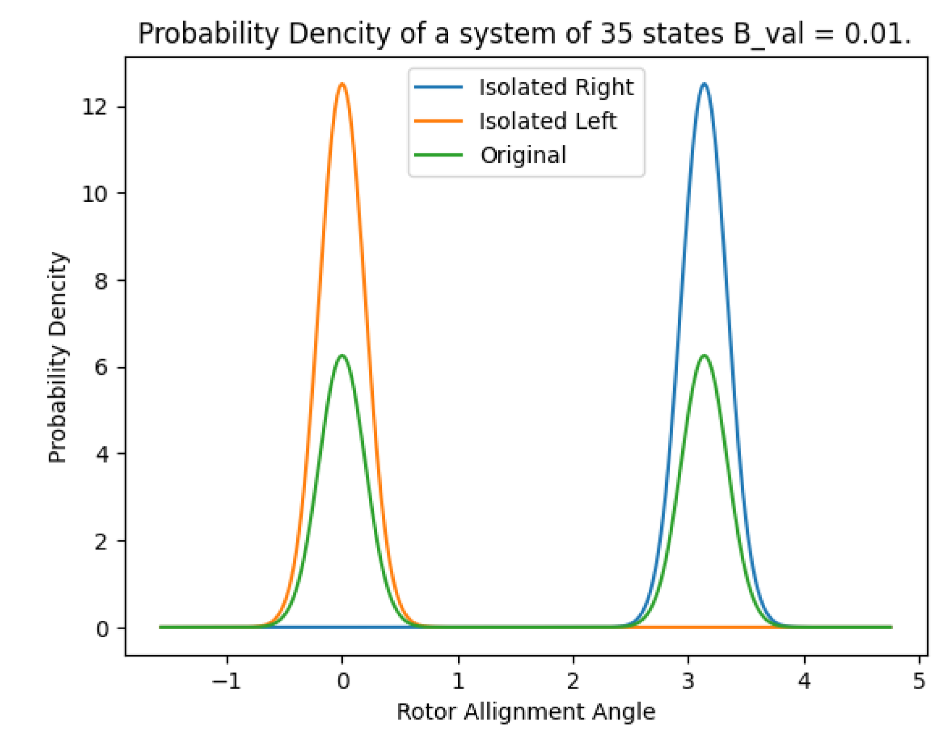


Figure 4: The example of probability dencityes separated onto localised states instead of superposition.

By solving the CCC equations for the system of Natural Orbital describing combined (isolated to the left or the right) state we can see a inverce prodoctional relationship between the value of B and the acuracy of CCC ground state energy.

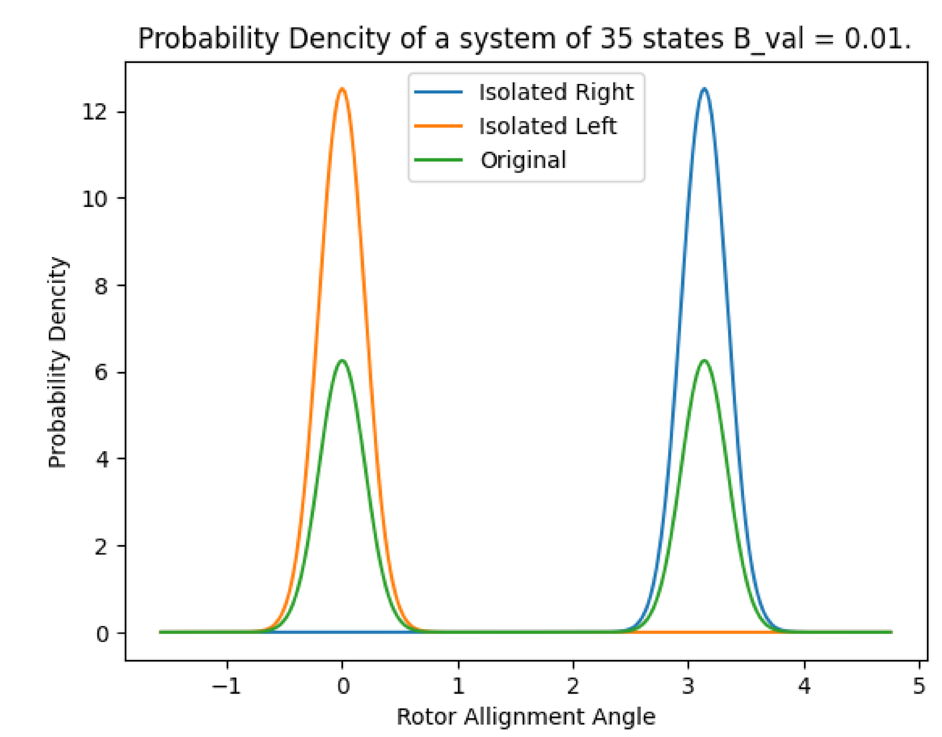


Figure 5: As B aproaches 0 (g aproaches infinity) acuracy increases.

Computational Efficency Results

Small $g < 100$ Limit

Future Work

References

Contact Information

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